

Teaching Robots to Feel: DAIMON's Massive Touch Dataset Aims to Transform Dexterous Manipulation

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Robots can see and hear, but they've largely been numb to the world until now. DAIMON Robotics, a Hong Kong-based startup, has unveiled **Daimon-Infinity**, a colossal dataset designed to give robot hands a sense of touch. Described as the largest **omni-modal dataset** for physical AI, it integrates high-resolution **tactile sensing** with vision, motion, and language across a vast range of tasks, from folding laundry to assembly-line manufacturing. At the heart of this initiative is DAIMON's advanced tactile sensor, a fingertip-sized module that packs over 110,000 effective sensing units. Unlike basic pressure sensors, this vision-based device captures **deformation**, **slip**, friction, material properties, and surface textures, enabling a comprehensive reconstruction of physical interactions. "Data scarcity remains a primary bottleneck in robot learning, particularly the lack of physical interaction data," said Prof. Michael Yu Wang, DAIMON's co-founder and chief scientist. To address this, the company built a distributed data-collection network that can generate millions of hours of data annually. They've open-sourced 10,000 hours of it to accelerate research. The dataset was developed with partners including Google DeepMind, Northwestern University, and the National University of Singapore. It spans over 80 real-world scenarios and 2,000 human-demonstrated skills, making it a rich resource for training general-purpose robotic foundation models. Prof. Wang, a pioneer in the field, advocates for a Vision-Tactile-Language-Action (VTLA) architecture, elevating touch to a modality on par with vision. "Our goal is to provide the high-quality 'fuel' needed to power embodied AI," he said. With this dataset, robots could soon perform delicate tasks that require a nuanced sense of touch from handling fragile items in warehouses to assisting in surgery. DAIMON envisions early deployments in hotels and convenience stores across China, where touch-enabled robots might soon handle everyday objects with human-like dexterity.

Word Treasure

omni-modal dataset -- A collection of data that captures multiple senses at once -- here, touch, vision, motion, and language -- to train robots more comprehensively.

tactile sensing -- The ability to detect pressure, texture, and temperature through physical contact, similar to the human sense of touch.

Daimon-Infinity -- The name of the enormous open-sourced dataset built by DAIMON Robotics to teach robots how physical interactions feel.

deformation -- The change in shape of an object (or sensor surface) when pressed or squeezed, providing key tactile information.

slip -- The small sliding movement between a robot's fingertip and an object; detecting slip is crucial for a stable grip.

VTLA architecture -- Short for Vision-Tactile-Language-Action, a design approach that gives touch equal status to seeing, reading, and moving in robot

learning.

Background you need

While robots have gained vision through cameras and speech through microphones, creating a high-resolution sense of touch has been a hard challenge because it requires sensors that can detect tiny forces, vibrations, and textures.

Prof. Michael Yu Wang, co-founder of Hong Kong-based DAIMON Robotics (a special administrative region of China known for its tech startup ecosystem), previously pioneered vision-based tactile sensors that use cameras inside a silicone finger to see deformation.

Major AI labs like Google DeepMind (the company behind AlphaGo and Gemini) and universities such as Northwestern University (a leading US institution in robotics) and the National University of Singapore have joined this project, reflecting a global race to build 'foundation models' for physical robots just as GPT models transformed language.

Data scarcity in physical interaction has long bottlenecked robot learning; open-sourcing 10,000 hours of omni-modal data aims to replicate the impact that ImageNet had on computer vision by giving researchers a massive, standardized touch dataset.

The vision-tactile-language-action (VTLA) architecture treats touch as a core sense, not an add-on, which could allow robots to handle everything from raw eggs to surgical tools with human-like dexterity.

Break it down

LEAD: Robots can see and hear but have been 'numb' -- until DAIMON's new dataset arrives

+ hook: 'described as the largest omni-modal dataset for physical AI'

KEY EVENT: DAIMON Robotics unveils Daimon-Infinity, a massive touch dataset for dexterous manipulation

TECHNICAL HEART: A fingertip-sized tactile sensor with over 110,000 sensing units

+ WHAT IT CAPTURES: deformation, slip, friction, material properties, surface textures

+ HOW: vision-based device, not just pressure sensing

THE PROBLEM: Data scarcity in physical interaction data -- quoted by Prof. Wang as a 'primary bottleneck'

+ SOLUTION: distributed data-collection network generating millions of hours annually

+ OPEN-SOURCE: 10,000 hours released to accelerate global research

COLLABORATORS: Google DeepMind, Northwestern University, National University of Singapore

+ SCOPE: Over 80 real-world scenarios and 2,000 human-demonstrated skills

VISION: VTLA architecture elevates touch to equal footing with vision and language

+ FUTURE DEPLOYMENTS: Hotels and convenience stores in China, maybe surgery and warehouses later

Why it matters

As robots move from factory cages into everyday human spaces, a refined sense of touch is what will stop accidents, enable delicate tasks, and make them genuinely useful; this dataset could fast-track that leap, influencing everything from healthcare to retail -- and shaping how future generations interact with machines.

Test what you remember

1. What kind of sense is DAIMON's dataset mainly designed to give robots?
 - (A) Hearing
 - (B) Taste
 - (C) Touch
 - (D) Smell
2. How many effective sensing units does DAIMON's fingertip-sized module pack?
 - (A) About 10,000
 - (B) Over 110,000
 - (C) Nearly 500,000
 - (D) Exactly 1 million
3. Who is Prof. Michael Yu Wang in the context of this project?
 - (A) The head of Google DeepMind's robotics division
 - (B) A professor at Northwestern University
 - (C) DAIMON's co-founder and chief scientist
 - (D) The lead author of the Ecology study
4. Which of the following is NOT mentioned as a partner in developing the dataset?
 - (A) Google DeepMind
 - (B) Northwestern University
 - (C) National University of Singapore
 - (D) Stanford University
5. How many hours of data have been open-sourced to researchers?
 - (A) 1,000 hours
 - (B) 10,000 hours
 - (C) 100,000 hours
 - (D) 1 million hours
6. Where does DAIMON envision early deployments of touch-enabled robots?
 - (A) In deep-sea exploration
 - (B) In outer space missions
 - (C) In hotels and convenience stores across China
 - (D) In hospitals across Europe

Answer key (try the quiz first!): 1: C 2: B 3: C 4: D 5: B 6: C

